1. Hardware

1.1. Computers and Their Components

- A general-purpose computer system comprises a processor, memory, and I/O functionality.
- The following essential features are needed in a computer -
  - **Input:** Takes in data from the outside world.
  - **Output:** Displays data for human understanding.
  - **Primary Storage:** Main memory storing critical program instructions and data.
  - **Secondary Storage:** Non-volatile storage for noncritical data.
  - **Removable secondary storage:**
    - File backup and archives
    - Portable transfer of files to a second device
- **Embedded systems:**
  - Miniature computer systems such as microprocessors that are often a part of a more extensive system.
  - Each embedded system performs a few specific functions, unlike general-purpose computers.

Benefits	Drawbacks
Reliable since there are no moving parts	Difficult to program functions since there is no interface
Require less power	Expensive expert help is needed for the repair
Cheap to mass-produce	

Principle Operations of Hardware Devices

Laser printer:

- A laser beam and rotating mirrors are used to draw an image of the page on a photosensitive drum
  - The image is converted into an electric charge, which attracts charged toner such that it sticks to the image
- Electrostatic-charged paper rolled against the drum
- Charge pulls toner away from drum and onto paper
- Heat applied in the fuser to fuse toner to the paper
- The electrical charge was removed from the drum, and excess toner was collected

please note: inkjet printers have been removed from the syllabus

3D Printer:

- The process starts with a saved digital file that holds the blueprint of the object to be printed
- The object is then built by sequentially adding layers of a material (e.g. polymer resin) until the object created
- The object is then cured (e.g. resin-made objects are hardened by UV light)

Microphone:

- Incoming sound waves enter the screen and cause vibrations in the diaphragm
- Vibrations cause the coil to move past a magnetic core
- Electrical current is generated, which is then digitized

Speaker:

- Takes electrical signals and translates them into physical vibrations to create sound waves
- The electric current in the voice coil generates an electromagnetic field
- Change in digital audio signal causes current direction to change, which changes field polarity
- Electromagnet is either attracted or repelled to a permanent magnet, causing a diaphragm that is attached to the coil to vibrate
- Vibration transmitted to air in front of the speaker
- The degree of vibration determines the amplitude and frequency of the sound wave produced

Magnetic Hard Disk:

- Hard disks have platters whose surfaces are covered with a magnetisable material.
- Platters are mounted on a central spindle and rotated at high speed
- The surface of platters is divided into concentric tracks & sectors, where data is encoded as magnetic patterns
- Each surface is accessed by read/write heads
- When writing, current variation in the head causes magnetic field variation on the disk
- When reading, magnetic field variation from the disk produces current variation in the read head

Solid State (Flash) Memory:

- Most use NAND-based flash memory
- Consist of a grid of columns & rows that has 2 transistors at each intersection
- Two transistors:
- Floating Gate: stores electrons, and the presence or absence of charge (electrons) represents either 1 or 0
- Control Gate: controls charge (electrons) flow for read/write

Optical Disc Reader/Writer:

- The disc surface has a reflective metal layer and is spun
- The tracking mechanism moves the laser assembly
- The lens focuses laser onto the disc
- A laser beam shone onto a disc to read/write
- Tracks have sequences of amorphous and crystalline states on the metallic layer
- When reading, the reflected light from the different states on the track is encoded as bit patterns
- When writing, the laser changes surface to crystalline and amorphous states along the track, corresponding to 1s or 0s.

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Touchscreen:

- Considered as both an input & output device
- There are two main-types:

Resistive	Capacitive
Consists of two charged plates	Made from materials that store electric charge
Pressure causes plates to touch, completing the circuit	When touched, the charge is transferred to the finger
Point of contact registered with coordinates used to calculate the position	

- Virtual (Reality) Headset:
  - Virtual headsets consist of 2 lenses, (an LCD) display, a circuit board with sensors, a cover and foam padding
  - The display provides a simulation of a 3D environment generated by a 3D graphics package
  - The user can 'move' in the virtual environment by moving their head or using controllers
- Buffers:
  - A queue that temporarily stores data to balance input/output speed of data, while the cache is the short-term memory storage that stores frequently used data.
- Random Access Memory vs. Read-Only Memory

RAM	ROM
Volatile memory: loses content when power is turned off	Non-volatile memory: does not lose content when power is turned off
It can be read and altered	It can only be read
Used to store currently executing program	Used for storing OS kernel and boot-up instructions

- Types of RAM - Static RAM vs. Dynamic RAM

SRAM	DRAM
Doesn't need to refresh; hence, it uses less power and faster access time	Has to be refreshed; it has slower access times and needs higher power
More complex circuitry, hence more expensive	Only a single transistor & capacitor, hence less expensive to purchase
Each bit is stored in a flip-flop	Each bit is stored as a charge
Has lower data density	Has higher data density
Used in cache memory	Used in main memory

- Types of ROM – PROM vs. EPROM vs. EEPROM

PROM	EPROM	EEPROM
Programmable ROM	Erasable Programmable ROM	Electrically Erasable Programmable ROM
It can be programmed only once after it is created	It can be erased by UV light exposure and can then be reprogrammed	It can be erased by an electrical signal and can then be reprogrammed
Data cannot be erased or deleted	Chip has to be removed for reprogramming	Can update data without removing the chip.

- Monitoring and Control Systems
  - Monitoring System:
    - Monitors some state external to the computer system
    - No changes were made to the environment by the system, and hence, no feedback
  - Control System:
    - Regulates the behaviour of other devices or systems
    - Event-driven system: the controller alters the system's state in response to some event
    - Time-driven system, where the controller takes action at a specific point in time
  - Hardware typically used in a system:
    - Sensor that measures an (analogue) property and transmits it to a processing unit, generally as an electrical or optical signal
    - Actuators that switch on/off heavy appliances (e.g. heater to heat/fan to cool)
    - ADC that converts analogue signals to digital signals
    - Transmission cable to transfer signals
  - Feedback Systems:
    - Output from the system affects the input of sensors
    - Ensures the system operates within the given criteria
    - Enabling the system output to affect subsequent system inputs may cause a change in the actions taken by the system
    - This enables the system to adjust conditions in a continuous process automatically

1.2. Logic Gates and Logic Circuits

- Logic Gates: use one or more inputs and produce a single logical output
- AND gate: If both inputs are high, the output is high ($A \cdot B$)

A	B	Output
0	0	0
0	1	0
1	0	0
1	1	1

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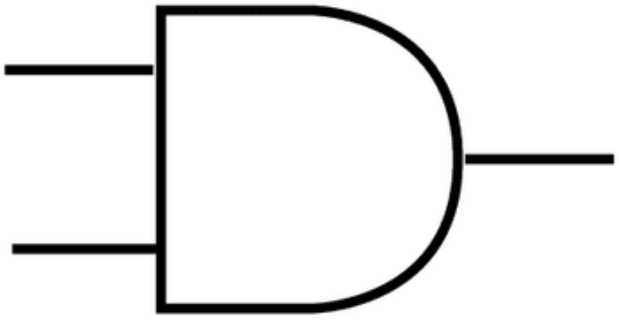
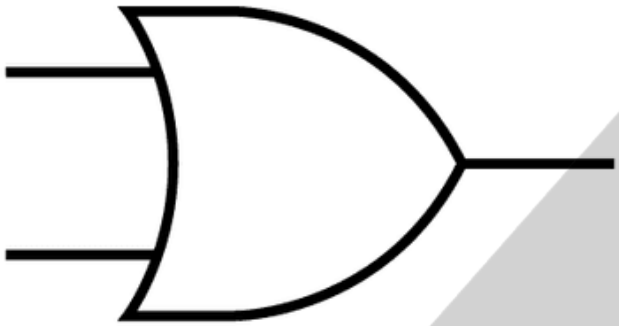


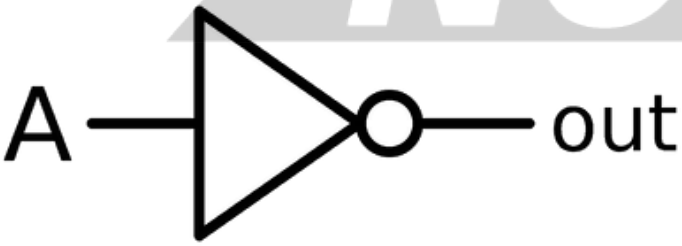
- OR gate: If either input is high, the output is high (*A+B*)

A	B	Output
0	0	0
0	1	1
1	0	1
1	1	1



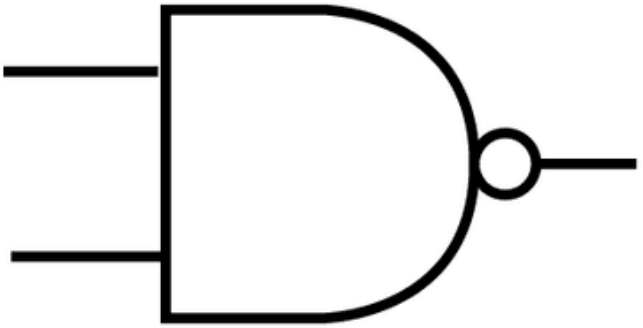
- NOT gate: an inverter (A)

A	Output
1	0
0	1



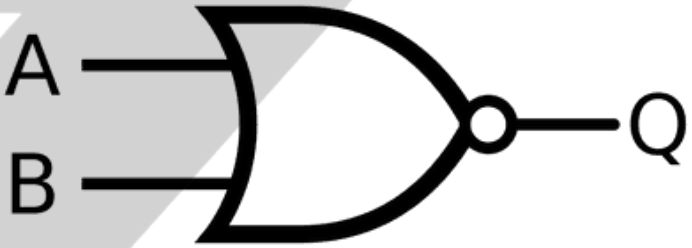
- NAND gate: (A•B)

A	B	Output
0	0	1
0	1	1
1	0	1
1	1	0



- NOR gate: (A+B)

A	B	Output
0	0	1
0	1	0
1	0	0
1	1	0



- XOR gate: ($A \oplus B$)

A	B	Output
0	0	0
0	1	1
1	0	1
1	1	0

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